Nonuniform Λ-Distributions on Cyclic Matrix Groups

Fourier weights, convolution walks, and direct determinant checks

Proof and computational verification note

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Outcome. The averaging-operator and convolution formulas were checked on explicit diagonal models of C_3, C_4, C_5 in dimensions $1, 2, 3$. Arbitrary nonuniform measures and four walk times were tested for every partition through total degree five. All **288** comparisons pass; the maximum absolute error is 2.91×10^{-14}. Three additional unexpanded determinant evaluations also pass.

1 The general coefficient theorem

Let G be finite, $\rho : G \to U(V)$, and let ν be a probability measure. For finitely many active variables define

$$\mathcal{S}_{\nu,V}(\mathbf{t}) = \sum_{g \in G} \nu(g) \prod_a \det(I - t_a \rho(g))^{-1}. \tag{1}$$

The standard identity $\det(I - tA)^{-1} = \sum_{r \geq 0} \chi_{\mathrm{Sym}^r V}(A) t^r$ implies, by multiplying the series variable by variable, that

$$[\mathbf{t}^\tau]\mathcal{S}_{\nu,V} = \sum_g \nu(g)\chi_{W_\tau}(g) = \mathrm{Tr}\left(\sum_g \nu(g)\rho_{W_\tau}(g)\right), \qquad W_\tau = \bigotimes_a \mathrm{Sym}^{\tau_a} V. \tag{2}$$

This proves the proposed replacement of the Haar invariant dimension by the trace of a nonuniform averaging operator. Positivity of the measure does not force the trace to be real or integral; the C_5 test below deliberately exhibits a complex coefficient.

If $\nu = \kappa^{*r}$ and the convention is that a step product is $X_1 \cdots X_r$, then representation multiplicativity gives

$$\widehat{\nu}(W) = \widehat{\kappa}(W)^r. \tag{3}$$

Equations (2)–(3) prove the convolution formula without a centrality assumption.

2 Specialization to C_m

Write $C_m = \{0, \ldots, m-1\}$ additively and $\zeta_m = e^{2\pi i/m}$. A diagonal representation is determined by weights $w_1, \ldots, w_d \pmod m$:

$$\rho(k) = \mathrm{diag}(\zeta_m^{kw_1}, \ldots, \zeta_m^{kw_d}). \tag{4}$$

Let $M_\tau(q)$ denote the multiplicity of the character $k \mapsto \zeta_m^{kq}$ in W_τ. Extracting weights from

$$\prod_{j=1}^{d} (1 - xz^{w_j})^{-1} = \sum_{r \geq 0} \sum_q M_{(r)}(q) x^r z^q \tag{5}$$

and convolving the multiplicities for the factors in W_τ gives the finite Fourier formula

$$[\mathbf{t}^\tau]\mathcal{S}_{\nu,V} = \sum_{q \bmod m} M_\tau(q)\widehat{\nu}(q), \qquad \widehat{\nu}(q) = \sum_{k=0}^{m-1} \nu(k)\zeta_m^{kq}. \tag{6}$$

For a walk this becomes

$$[\mathbf{t}^\tau]\mathcal{S}_{\kappa^{*r},V} = \sum_q M_\tau(q)\widehat{\kappa}(q)^r. \tag{7}$$

Thus the abstract matrix Fourier transform reduces to a transparent scalar filter on each weight.

3 Independent verification routes

The direct route builds every matrix in (4), computes the power traces $p_j = \mathrm{Tr}(\rho(k)^j)$, and obtains complete characters from Newton's recurrence

$$h_0 = 1, \qquad h_r = \frac{1}{r}\sum_{j=1}^{r} p_j h_{r-j}. \tag{8}$$

It then averages $\prod_a h_{\tau_a}$ using the fully enumerated probability distribution. The formula route never uses matrix traces: it constructs the integer weight multiplicities in (5) and applies (6) or (7). For walks, the direct distribution is obtained by repeated cyclic convolution, whereas the formula route raises Fourier scalars to the rth power.

As a full target check, the code also evaluates (1) at $(t_1, t_2) = (0.07, -0.04)$ using matrix determinants and compares it with the product over the weights in (4). These values stay safely inside every pole.

4 Cases and results

The representations and the nonuniform arbitrary measure are

Group	weights	dimension	$\nu(k)$ before normalization
C_3	(1)	1	$1, 2, 3$
C_4	$(1, -1)$	2	$1, 2, 3, 4$
C_5	$(0, 1, 2)$	3	$1, 2, 3, 4, 5$

The walk step is $0.35\delta_0 + 0.40\delta_1 + 0.25\delta_{-1}$ and is tested at $r = 0, 1, 2, 5$. There are 19 partitions through total degree five, hence $3 \cdot (1 + 4) \cdot 19 = 285$ coefficient comparisons plus three determinant comparisons.

Case	coefficient	direct value	formula value	result
C_3, arbitrary	$\tau = (3)$	1.0000000000	1.0000000000	**PASS**
C_4, walk $r = 5$	$\tau = (2, 1)$	-0.0210525000	-0.0210525000	**PASS**
C_5, arbitrary	$\tau = (2, 2)$	$2.1666667 + 0.2293970i$	$2.1666667 + 0.2293970i$	**PASS**

The complex C_5 row is a useful diagnostic: the implementation is genuinely testing non-Haar averaging, not silently rounding to invariant dimensions.

5 Reproducibility and scope

Run `cyclic_groups/verification.py`, or open `cyclic_groups/notebook.py` in marimo. The notebook exposes the degree cutoff and every comparison. The finite computations verify all coefficients in the stated range; equations (1)–(7) are all-degree proofs. No sampling is used.